

***D*-plus Version 1 User Manual**

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Citation: When you use *D*-plus, please cite our corresponding manuscript: *D-plus: Beyond the D-statistic — Accurate Detection and Characterization of Introgression Despite Among-Lineage Rate Variation*.

License and Attribution: This software builds upon the source code of the *3s* program (Yang 2010; Zhu and Yang 2012; Dalquen et al. 2017) and contains substantial custom modifications. It is distributed under the GNU General Public License v3.0. Please consult the included LICENSE file for full license terms.

Table of Contents

1. Introduction
2. Compilation & Dependencies
3. Input-data Formats
 - 3.1 Multilocus sequence alignment file
 - 3.2 Control-file parameters
4. Program execution modes ([*run*](#) parameter)
5. Model specification syntax
6. Output files and interpretation
7. References

1. Introduction

D-plus is a **maximum-likelihood phylogenomic tool designed as a downstream companion to the *D*-statistic (ABBA-BABA test)** (Green et al. 2010). The traditional *D*-statistic can yield significant results not only from non-sister introgression (the default interpretation), but also from lineage-specific substitution-rate heterogeneity (Frankel and Ané 2023; Koppetsch et al. 2024; Pang et al. 2025) and introgression from unsampled lineages (Tricou et al. 2022; Pang and Zhang 2024) — competing sources that the test alone cannot discriminate. *D*-plus explicitly accommodates both rate variation as and multiple type of introgression (non-sister introgression and ghost introgression). It uses sequential likelihood-ratio tests (LRTs) to:

1. Distinguish spurious *D*-statistic signals caused by substitution-rate heterogeneity from genuine introgression;
2. Classify genuine introgression signals into ghost introgression, non-sister introgression, or their joint occurrence;
3. Return maximum-likelihood estimates of key evolutionary parameters: introgression probabilities, introgression time, divergence times, and relative substitution rate.

D-plus operates on the three ingroup taxa of a *D*-statistic quartet with the topology ((P1, P2), P3). It takes multi-locus nuclear sequence alignments as input and implements Likelihood Rate Test (LRT).

Limitation of version 1:

1. Only one haploid sample per species is supported.
2. Introgression between sister lineages is not considered. Only seven predefined introgression models (M1-M7) are configurable (see below).
3. Ambiguous IUPAC nucleotide codes (e.g., N, R, Y) cannot be handled. Any alignment site containing ambiguous characters in any of the three input sequences will be discarded automatically.

2. Compiling and Running the Program

Dependencies

D-plus requires the **GNU Scientific Library (GSL)** and an OpenMP-enabled C compiler (e.g., GCC, Clang). Install GSL on your system before compilation.

```
# set install prefix inside your home folder
export GSL_PREFIX=$HOME/software/gsl-2.8
mkdir -p $GSL_PREFIX

# download source tarball
wget https://ftp.gnu.org/gnu/gsl/gsl-2.8.tar.gz
tar -zxvf gsl-2.8.tar.gz
cd gsl-2.8

# configure: install to your home directory, NO sudo required
./configure --prefix=${GSL_PREFIX}

# compile and install
make -j$(nproc)
make install
```

Compilation command

Replace `PATH_GSL` with your actual GSL installation prefix path (e.g., `/usr/local`).

```
cc -o D-plus -O3 -std=c99 -I PATH_GSL/include -L PATH_GSL/lib D-plus.c -
fopenmp -lm -lgsl -lgslcblas
```

- `-O3`: high-level optimisation; - `-fopenmp`: enables multi-thread parallel computing;
- `-lgsl -lgslcblas`: links against the GSL numerical-library routines.

Successful compilation generates executable binary **D-plus**.

Program invocation

D-plus reads all runtime settings from a plain-text **control file**. Launch from shell:

```
./D-plus my_control.ctl
```

3. Input-data formats

3.1 Sequence data file

Input sequence file follows the **multi-locus PHYLIP format**. All loci are stored sequentially within one single file.

Each sequence identifier must contain a species label after the symbol ^. This species labels must exactly match the three names defined by the **stree** entry in the control file (see section 3.2).

Example fragment:

```
3 34
loci1^A CGACCAGGACGTCACCTCGTATAAGAAGCCTATAC
loci1^B AGACCAGGACGTCACCTCGTATAAGAAGCCTATAC
loci1^C AGACCAGGACGTCACCTCCTATAAGAAGCCTATAC
3 20
loci2^C TTCTAACAGCTGCGAGATAA
loci2^B TTCTAACAGCTGCGAGATAA
loci2^A TTCTAACTGCTGCGAGATAA
```

Any alignment site containing gap characters (-) or ambiguous IUPAC codes (N, R, Y etc.) across any of the three sequences is automatically discarded for that locus.

3.2 Control-file format

The control file is plain text with key = value syntax. A line is treated as a comment if its first non-whitespace character is *.

Example excerpt of a control file:

```
jobname = test
seqfile = fsc.1.phy
stree = A B C
nloci = 1000
ABBA = 0
RV = 1
run = 2
repeat = 3
*model = U 2 3
npoints = 16
nthreads = 10
```

Parameter interpretation:

Parameter	Description
jobname = test	Required; Base name for output files. Main output: test.out.
seqfile = 1.phy	Required; File path to your multi-locus sequence input file.
stree = A B C	Required; Defines the three ingroup species for topology ((P1, P2), P3). The order is strictly P1 P2 P3. Critically, the ordering of P1 and P2 should be chosen such that it yields a positive <i>D</i> -statistic. Species labels must exactly match labels after ^ inside your sequence file.
nloci = 1000	Number of loci used for analysis; program reads the first <i>nloci</i> locus blocks from your sequence file. If this parameter is omitted from the control file, all available loci in the input sequence file will be used by default.
ABBA = 0	When ABBA = 1, <i>D</i> -plus tallies counts for 3 site patterns: ABBA, BABA and BBAA. Each locus is treated as fully linked; therefore, at most one informative site from a locus contributes to the site-pattern counts. This output is generated only when the PHYLIP file includes an outgroup with the exact species label “Outgroup”. If ABBA < BABA, <i>D</i> -plus automatically swaps P1 and P2. Default = 0.
RV = 1	RV = 1 (default): enable relative-rate heterogeneity between sister lineages for tree model M0 and introgression models M1-M7. RV = 0: enforce strict molecular clock.
run = 2	Execution-mode switch; see Section 4 for full description. Default = 2.
repeat = 3	Number of independent optimization repeats for maximum likelihood search; multiple repeats reduce risk of convergence to local likelihood optima. Default = 3.
model = B 2 3	Required only when run = 0 or 1; Specifies introgression configuration. See Section 5.
npoints = 16	Number of Gaussian-quadrature integration points for coalescent integrals. Default = 16. Larger values improve numerical accuracy but increase runtime.
nthreads = 10	Number of OpenMP parallel threads for per-locus likelihood computation. Set according to your CPU-core count. Default = 1.

4. Program execution modes (*run* parameter)

- **run = 0** Model-fitting only, no hypothesis testing. Fit the user-specified introgression model given by *model* = ... and return MLE parameter estimates.
- **run = 1** Single pairwise LRT test: compare null tree model (M_0) against one user-specified introgression model defined by *model* = This follows the original logic of the *3s* software for introgression detection.
- **run = 2** (default empirical-analysis mode) Perform complete sequential LRT pipeline (Fig. 1) for decomposing the sources of significant *D*-statistic signals.

***D*-plus workflow for disentangling sources of significant *D*-signals**

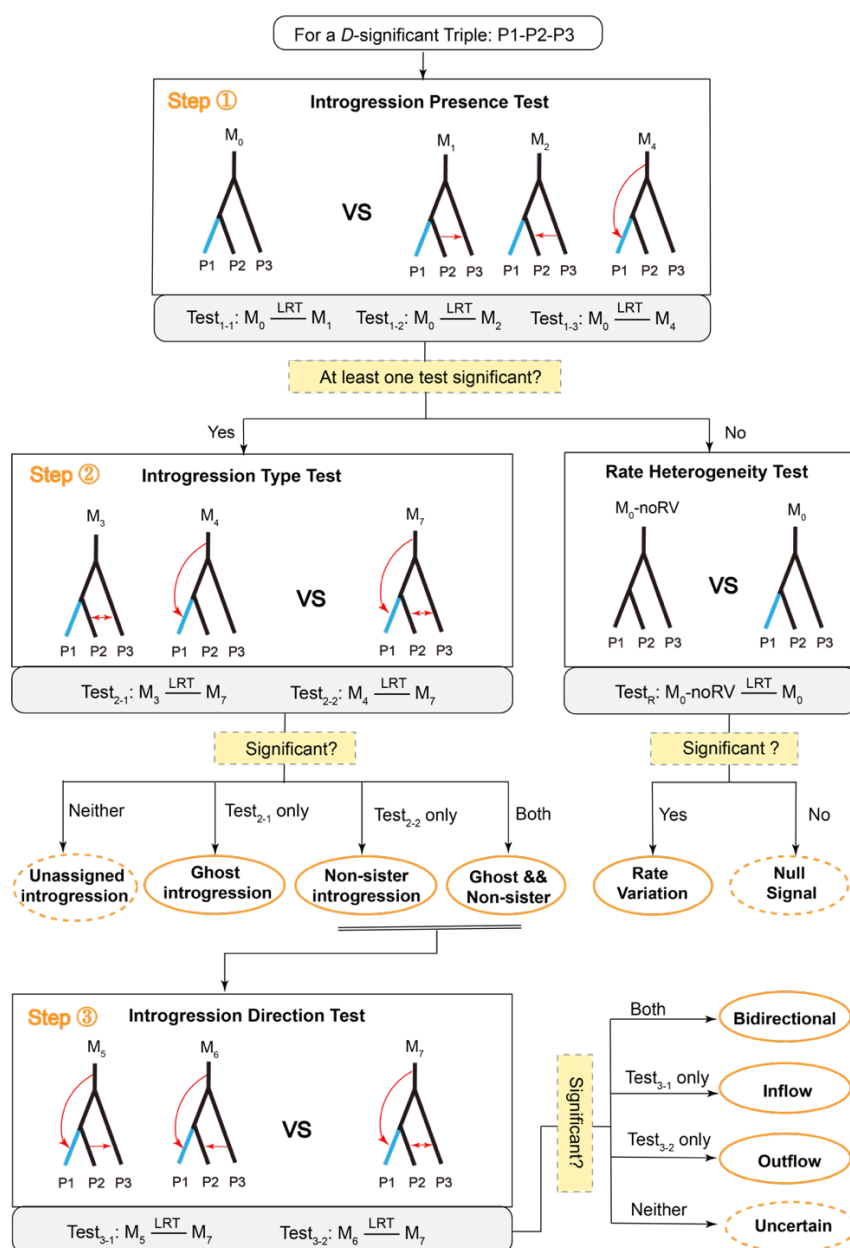


Figure 1. Three-stage sequential testing procedure of *D*-plus

5. Model-specification syntax (*model* parameter)

Only active under $\text{run} = 0$ and $\text{run} = 1$. Note that only 7 distinct introgression models (M1-M7, Fig. 2) are configurable in the current implementation.

Event-type codes:

- U: unidirectional non-sister introgression
- B: bidirectional non-sister introgression
- G: ghost-lineage introgression

Taxon numeric codes:

- 0: unsampled ghost lineage
- 1: P1 (first species in stree)
- 2: P2 (second species in stree)
- 3: P3 (third species in stree)

Syntax: `model = EVENT Donor Recipient [ & EVENT Donor Recipient ]`

Examples:

1. `model = B 2 3` Bidirectional introgression between P2 (donor = 2) and P3 (recipient = 3), corresponds to model M3;
2. `model = G 0 1` Ghost introgression: introgression from ghost lineage 0 into P1 (taxon 1), corresponding to model M4;
3. `model = U 2 3 & G 0 1` Combined model: unidirectional non-sister introgression $P2 \rightarrow P3$ plus ghost $\rightarrow P1$ introgression (corresponds to model M5).

Multiple events are separated by `&`.

Note: The sequential-testing pipeline ($\text{run} = 2$) internally cycles over all predefined models M0-M7; you do not supply the model keyword when using $\text{run} = 2$.

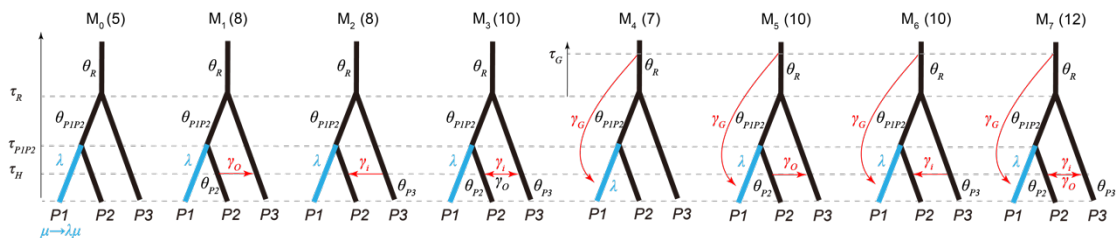


Figure 2. Tree model M0 and all configurable introgression models M1–M7 implemented in *D-plus*

6. Output files and interpretation

The main output file is `{jobname}.out`. Output content differs substantially depending on the execution mode selected via the *run* parameter.

Output components across all run modes

1. Echo of parsed control-file settings: prints every key-value pair loaded from the control file.
2. Fitted model block for each evaluated model:
 - Model identifier and brief model description.
 - Maximum log-likelihood value (lnL).
 - Table of maximum-likelihood estimates (MLEs) for all free parameters of that model.
 - Computation time consumed for this model optimisation.
3. LRT blocks (only generated when hypothesis testing is performed):
 - LRT statistic $2\Delta\ln L$ (twice the log-likelihood difference between alternative and null model).
 - Pre-computed critical value of the asymptotic χ^2 distribution at the 1 % significance level.
 - Significance flag: ns = non-significant; ** = significant at $\alpha = 0.01$.
 - Conclusion for each LRT comparison.

• Output for *run* = 0 (Model-fitting only, no hypothesis testing)

Only the user-specified introgression model defined by the *model* keyword is optimized. Parameter naming convention in output:

- tau_R: root divergence time of the P3 lineage
- tau_P1P2: divergence time of the P1-P2 ancestral node
- tau_H: introgression time for non-sister introgression
- tau_G: divergence time of ghost lineage (ghost-introgression models only for M4-M7)
- lambda_P1: relative substitution rate of P1 lineage
- phi_i: inflow introgression probability ($P3 \rightarrow P2$)
- phi_o: outflow introgression probability ($P2 \rightarrow P3$)
- phi_g: ghost-introgression probability

Example excerpt:

Parsed ctl file:

```

outfile = fsc.1.out
seqfile = fsc.1.phy
stree   = A B C
repeat = 1
model = M3
nloci = 1000

### Model 3 (Inflow && Outflow)
lnL = -92685.064990
MLEs:
tau_R      tau_P1P2    tau_H      phi_i      phi_o      lambda_P1
0.005038   0.002476    0.002476   0.002850   0.000011   1.976346
Time used: 0:16

```

- **Output for *run* = 1 (Single pairwise LRT)**

Performs one nested-model comparison: null tree model **M0** versus the **user-provided introgression model**. Both null and alternative models are optimized.

Example excerpt:

Parsed ctl file:

```

outfile = fsc.1.out
seqfile = fsc.1.phy
stree   = A B C
repeat  = 1
model   = M3
nloci   = 1000

```

```

### Model 0 (Species Tree)

```

```

lnL = -92685.115489

```

```

MLEs:

```

tau_R	tau_P1P2	lambda_P1
0.005042	0.002345	2.029481

```

Time used: 0:02

```

```

### Model 3 (Inflow && Outflow)

```

```

lnL = -92685.064864

```

MLEs:

tau_R	tau_P1P2	tau_H	phi_i	phi_o	lambda_P1
0.005038	0.002476	0.002476	0.002850	0.000010	1.976354

Time used: 0:14

==== Pairwise LRT: M0 (null) VS M3 (introgression) =====

2DlnL (M3 vs M0) = 0.101249 < chi2CV_1pct[5] = 15.086300 ns

LRT non-significant ($p \geq 0.01$): cannot reject null model M0.

- **Output for *run* = 2 (Sequential LRT pipeline, default empirical-analysis mode)**

Executes the complete three-stage workflow in Figure 1.

Example excerpt:

Parsed ctl file:

outfile = fsc.1.out

seqfile = fsc.1.phy

stree = A B C

repeat = 1

nloci = 1000

Model 0 (Species Tree)

lnL = -92685.106004

MLEs:

tau_R	tau_P1P2	lambda_P1
0.005038	0.002369	2.020306

Time used: 0:02

Model 1 (Outflow)

lnL = -92685.107505

MLEs:

tau_R	tau_P1P2	tau_H	phi_o	lambda_P1
0.005038	0.002369	0.001992	0.000010	2.020290

Time used: 0:11

Model 2 (Inflow)

Significant Rate Variation between sister lineages detected ($p < 0.01$) (M0)

7. References.

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